

5.3.4

AI25BTECH11010 - Dhanush Kumar

Question

Solve the following equations for x and y using cross-multiplication method

$$(ax - by) + (a + 4b) = 0$$

$$(bx + ay) + (b - 4a) = 0$$

Solution

The system of equations can be written in vector-matrix form as:

$$\begin{pmatrix} a \\ -b \end{pmatrix}^T \mathbf{X} = -a - 4b \quad (1)$$

$$\begin{pmatrix} b \\ a \end{pmatrix}^T \mathbf{X} = 4a - b \quad (2)$$

Using Cramer's Rule (matrix method), which is algebraically equal to the cross-multiplication method, the solution is obtained as follows.

The coefficient matrix and its determinant are:

$$A = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \quad (3)$$

$$\det(A) = a^2 + b^2 \quad (4)$$

The determinant for the first component of $\mathbf{X}$ is obtained by replacing the first column with the constants:

$$A_x = \begin{pmatrix} -a - 4b & -b \\ 4a - b & a \end{pmatrix} \quad (5)$$

$$\det(A_x) = (-a - 4b)a - (-b)(4a - b) \quad (6)$$

$$= -a^2 - 4ab + 4ab - b^2 \quad (7)$$

$$= -(a^2 + b^2) \quad (8)$$

The determinant for the second component of $\mathbf{X}$ is obtained by replacing the second column with the constants:

$$A_y = \begin{pmatrix} a & -a - 4b \\ b & 4a - b \end{pmatrix} \quad (9)$$

$$\det(A_y) = a(4a - b) - b(-a - 4b) \quad (10)$$

$$= 4a^2 - ab + ab + 4b^2 \quad (11)$$

$$= 4(a^2 + b^2) \quad (12)$$

Finally, using Cramer's Rule, the solution vector is:

$$\mathbf{X} = \frac{1}{\det(A)} \begin{pmatrix} \det(A_x) \\ \det(A_y) \end{pmatrix} \quad (13)$$

$$\mathbf{X} = \begin{pmatrix} -1 \\ 4 \end{pmatrix} \quad (14)$$

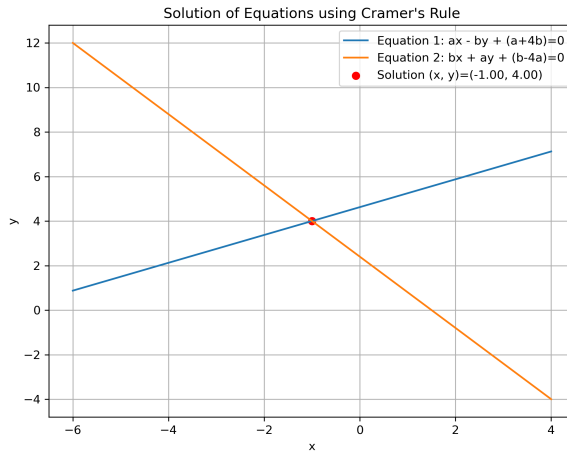


Fig. 0